



An Introduction to Game design and level Design

What does it mean to design a level?

- You have to understand the player's capabilities
 - Mechanics / actions available to player
 - Assumption of player skill level
- You are laying out the game geography
 - Location and interaction of challenges
 - Dynamic features (NPCs)
- You are also determining player progression
 - How does the player progress through the game

focus on each of these...

- 1. Focus on the player's capabilities
 - That's when we train the player to understand the game's systems and improve their own skills
- 2. Focus on player progression
 - That's when we focus on storytelling (or the experience overall)
- 3. Focus on game geography
 - That's when we come up with the actual layout

1. Teaching the Game

- A player coming into your game falls into one of these categories:
 - Has no idea how to play the game or any game of this type (complete novice)
 - Has never played your specific game, but knows the conventions of the genre (some knowledge)
 - Has played similar games or prequels (domain knowledge)
 - Has played this specific game before (expert)

1. Teaching the Game - Novice

- Someone who has never played your game, nor any other game in the genre
- Well, they could read the manual...
 - Yeah, no. That's the best way to turn off a player
 - (Back in the day, manuals mattered... but not really to learn how to play...)
- Let's make some tutorial levels!
 - Start playing immediately
 - AND learn while doing it!

The Tutorial Level

- Start with your design doc
 - Remove everything except the simplest core mechanic
 - Disable numerous actions and interactions
 - “Dumb down” your game
- Add mechanics back in one at a time
- You don’t HAVE to add a new mechanic for EACH level
 - Some take time. This can affect your layouts.

Conversation

Lets talk about some games introduction level and how they work.

PSA: Try to avoid this:



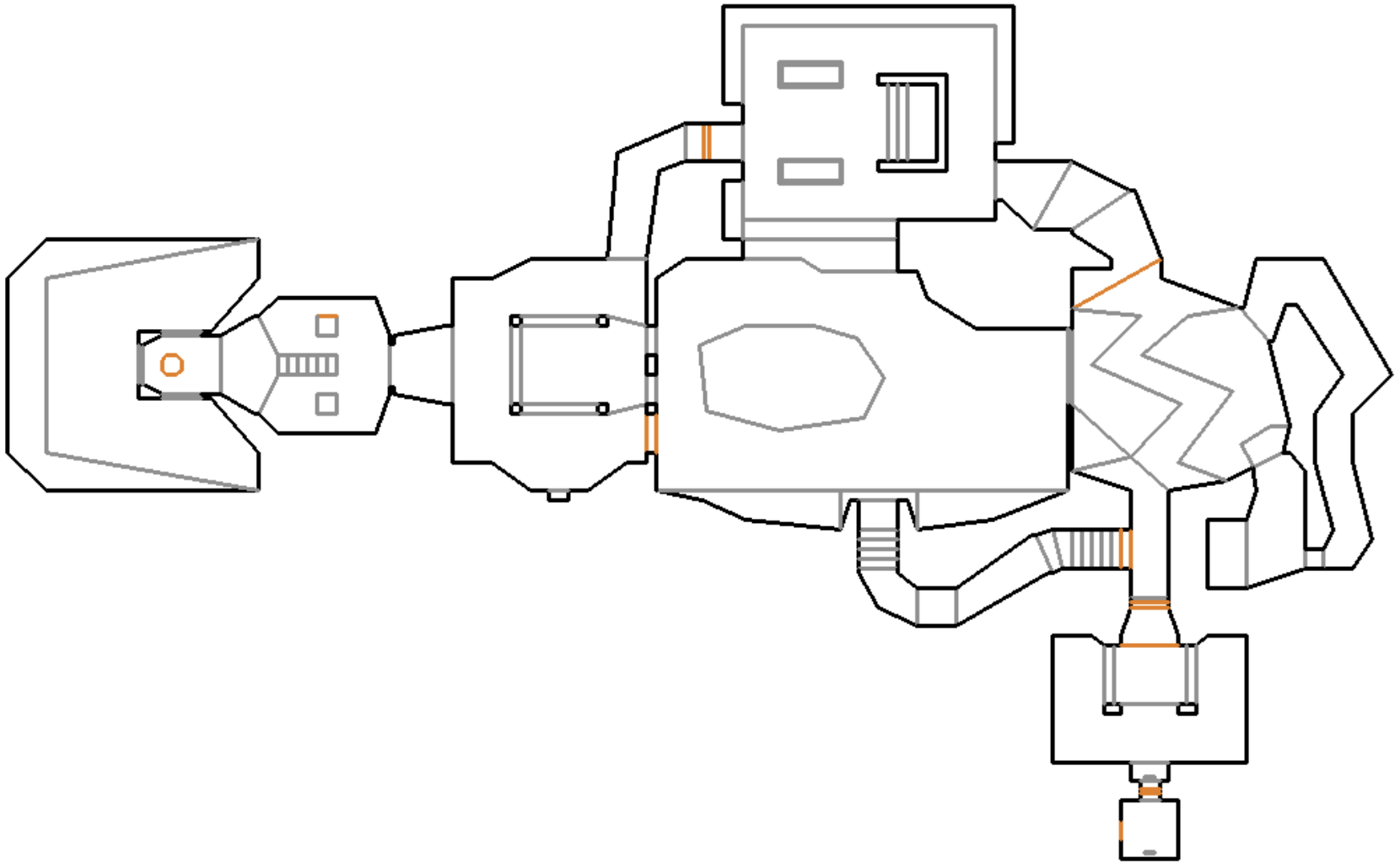
2. Focus on Player Progression

- Two schools of thought here:
 - Start very small and build up
 - Give everything and let the user get a taste of what's coming
- “Finish the first level last.” – John Romero, creator of Doom
- Another option: put a lot of content in the tutorial level to encourage replay with different ideas

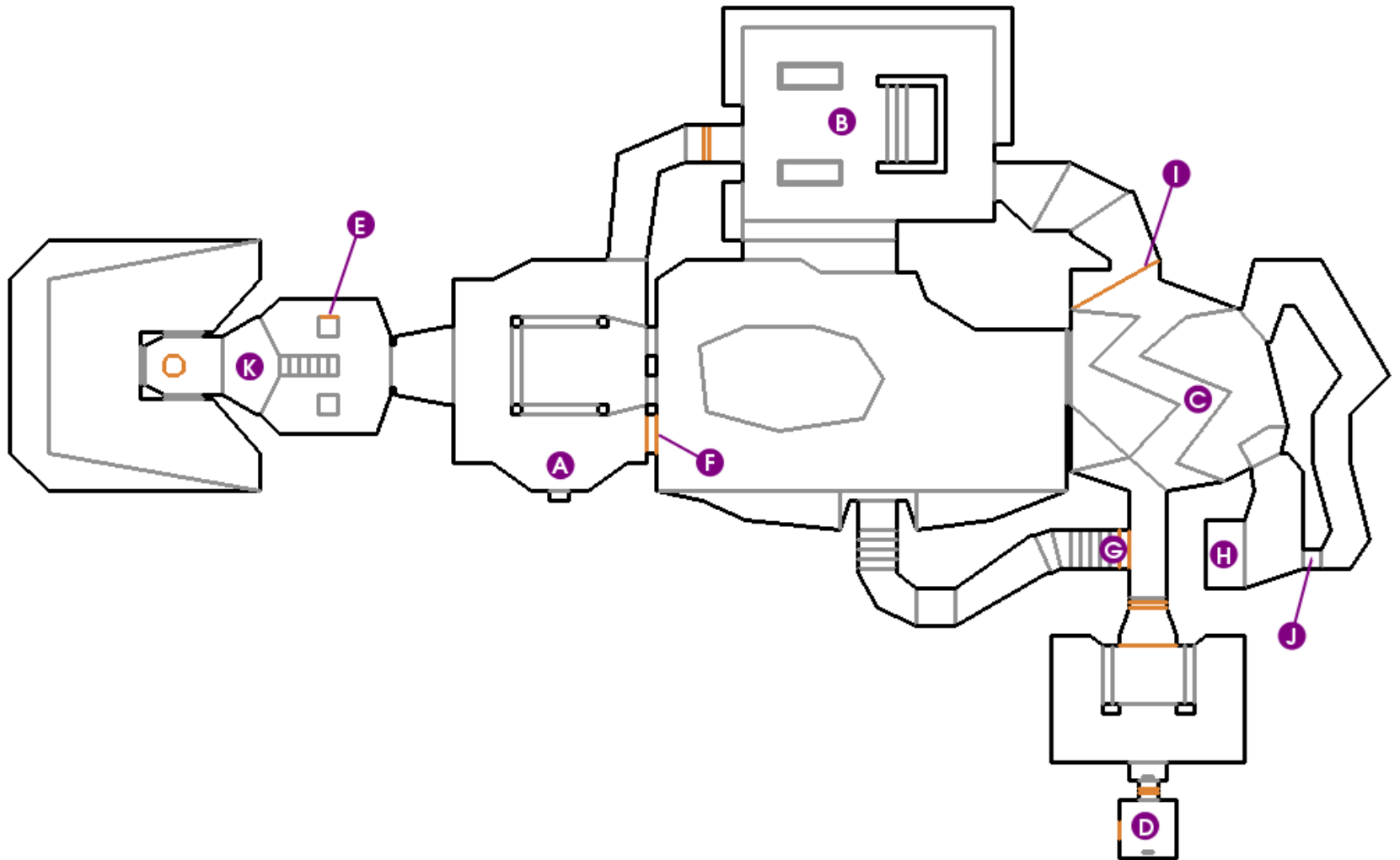
Doom Hangar

- Put everything in first level. So that your player will have some ways to understand the mechanics before the real challenge begins.

The Hanger - Doom



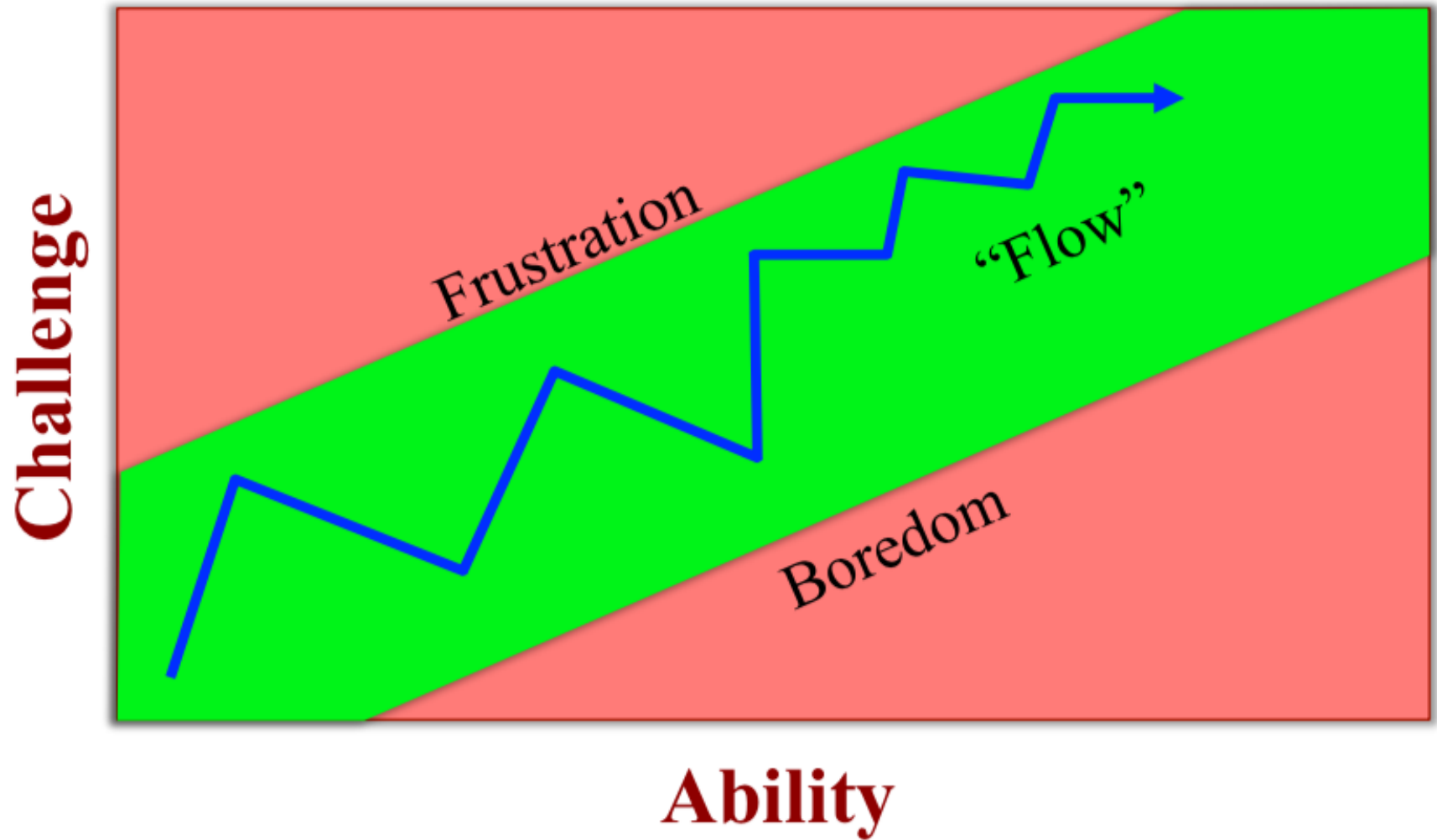
The Hanger - Doom



Upgrading Mechanics

- How do we introduce new mechanics one at a time?
 - Celeste
- Often treated as a form of training
 - “Master this thing and then you can have another thing”
- Bad end to this: grinding

Training and Flow



Enabling Flow

- Challenging activity that requires some skill
 - Physical, mental, or social skill
 - Without skill => Cannot do it
- Goals and Feedback
 - Player has to know what they are working toward
 - Constant feedback as to whether the goal is being approached
 - Have to know if success or failure is achieved (clear win/fail state)

Designing a Tutorial Level

- Identify the learning objective
 - What should the player learn?
 - Not necessarily the same as beating the level
- Identify player assumptions
 - What should the player know how to do now with the mechanics?
 - How much skill should the player have?
- Storyboard the progress
- Note: Puzzle design is the inverse of this

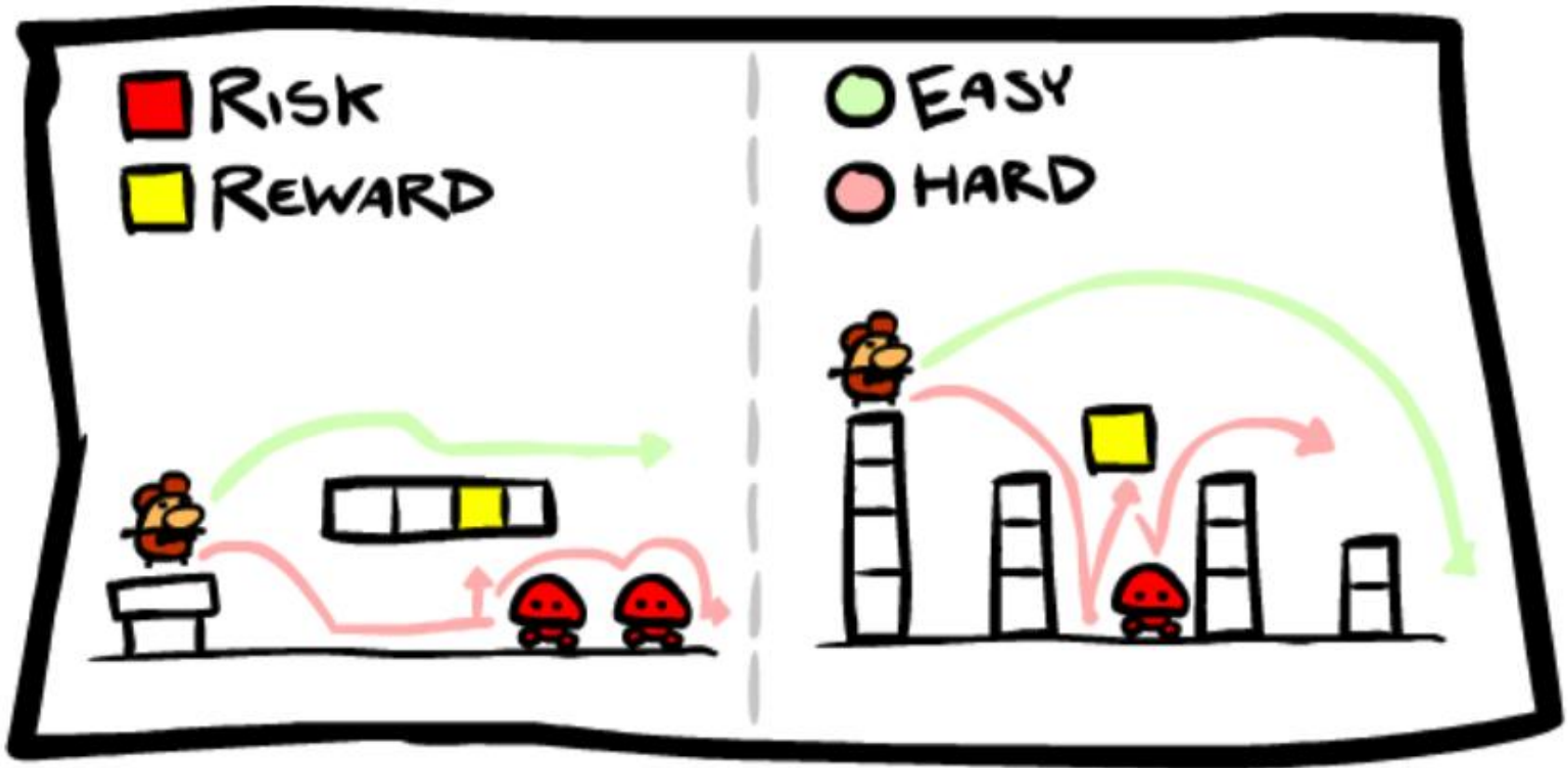
Mega Man

- Early Mega Man games do a great job of teaching the mechanics of a level and offering this progression
- Let's check it out:
- <https://www.youtube.com/watch?v=TI8fmQDFIfA>

3. Focus on Game Geography

- Games are all about “meaningful choice”
- Players need to be able to “own” their playthrough of the game
- Choosing a new weapon that +1 better than something with power of 530 already isn’t choice
- Choosing to “go high” or “go low” in a level IS choice

Analysis Paralysis – Risk and Reward



[Edmund McMillen, edmundm.com]

Build up the Level

- Design individual challenges
 - Maybe not quite “set pieces”, but similar
 - Choose a single obstacle or mechanic or NPC
- Figure out where in the level this might occur
 - This could be a single frame in a storyboard
- Combine pieces to make a full level

Super Mario Bros. 3

- Level 1-1
 - <http://www.mariouniverse.com/images/maps/nes/smb3/1-1.png>
- Level 1-5
 - <http://www.mariouniverse.com/images/maps/nes/smb3/1-5.png>

Donkey Kong Country Tropical Freeze

- Windmill Hills
- <https://www.youtube.com/watch?v=deuxJxOaYfc>

What About Open World Games



BOTW: Tutorial

- First: Tutorial.
 - Placed on a small plateau high up
 - Item you need to get off (paraglider) not given to you yet
 - You need to learn the game loop:
 - Activate tower
 - Find shrine (3 or 4)
 - Navigate environment to reach shrine
 - Solve small puzzle inside shrine
 - Power up (more life, more stamina, etc.)

BOTW: Some Notes

- A couple things that help limit cognitive overload:
 - Map is empty. Must reach a tall tower to unveil bits of map at a time.
 - These towers are very tall so they are often first destination when heading into new area
 - Once map part revealed, interesting things in map stick out and nudge you.

BOTW: Some Notes cont'd

- Two other things.
 - Game only gives you two quests (basically). Beat Ganon at the castle, and Free the divine beasts (optional)
 - There is one more though. Go to Kakariko Village and talk to IMPA. This is the only time this game explicitly asks you to go to some location
 - Game is guiding where it wants you to go, but will let you rebel if you'd like.

BOTW: After Tutorial

- A couple interesting design things here.
- Overall approach: soft guidance
 - 1. Visual landmarks visible from high altitudes
 - 2. Harsh weather prevents certain travel paths
 - 3. Difficult enemies restrict some paths
 - 4. Cutscenes to push player towards major goals

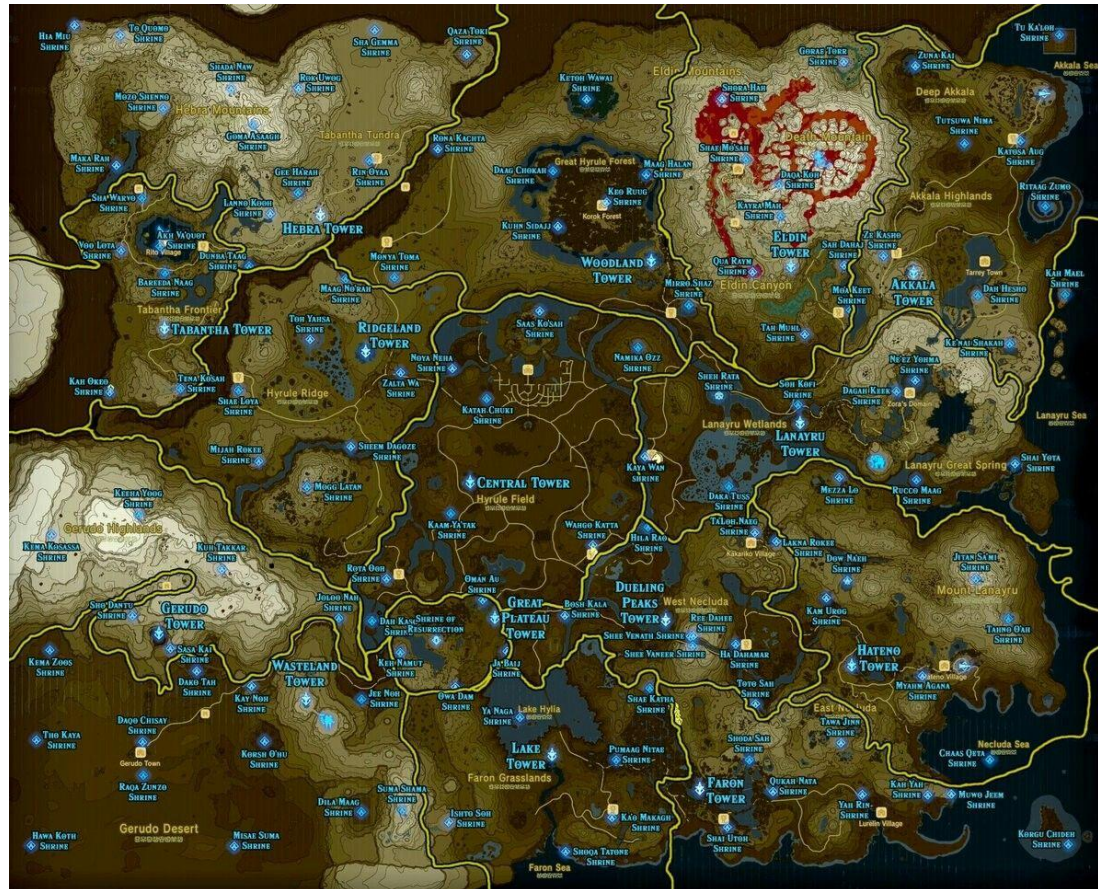
BOTW

1. Visual landmarks visible from high up



BOTW

2. Harsh weather prevents some paths



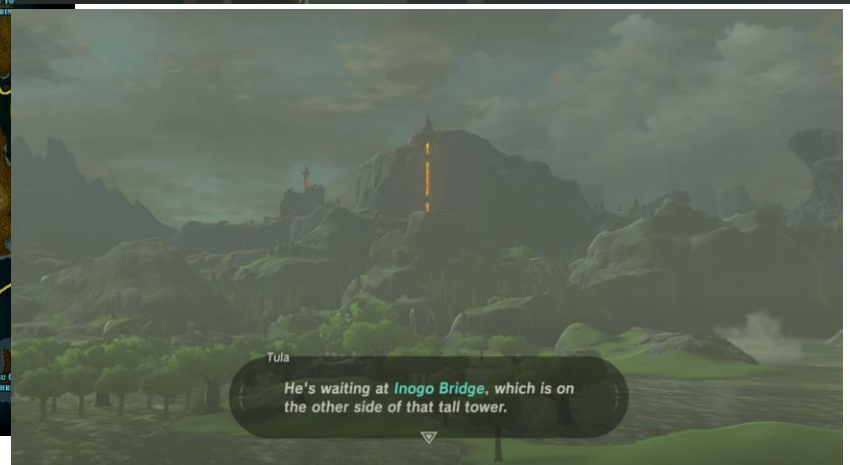
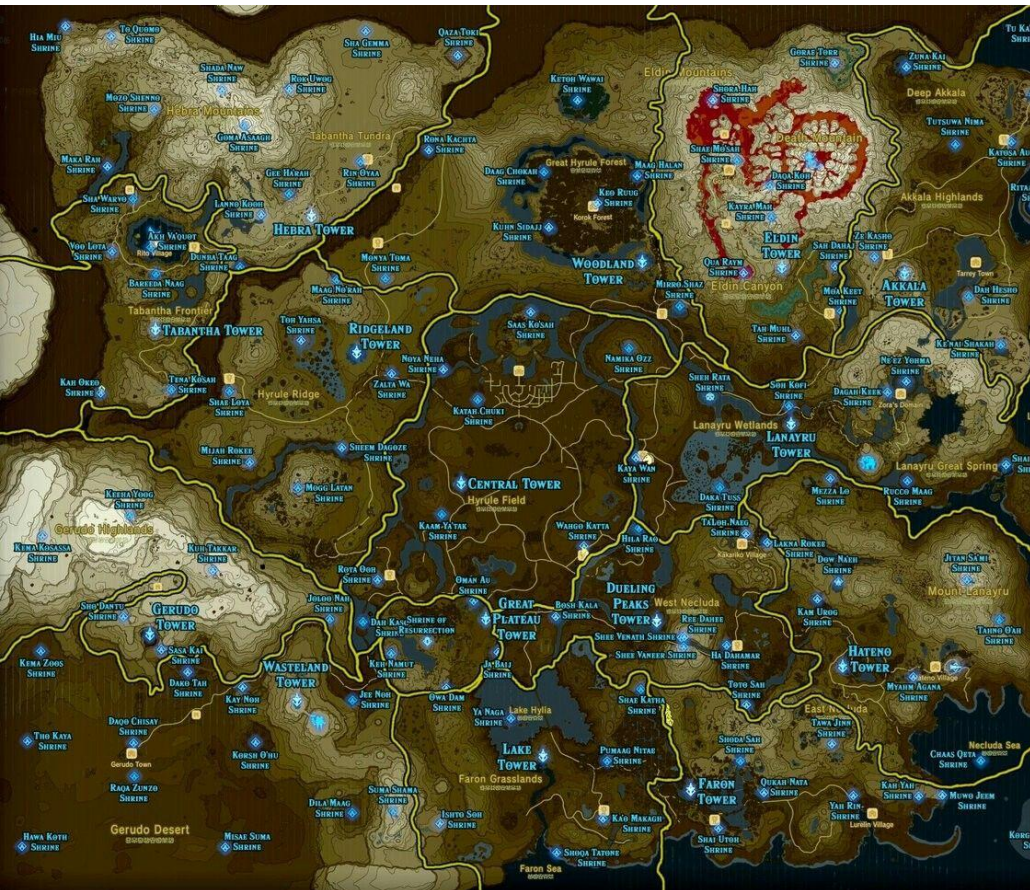
BOTW

3. Difficult enemies restrict some paths



BOTW

4. Cutscreens to push player towards major goal



Summary

- Level Design:
 - Think about **players, tutorials, and how to guide**. Could be soft guidance, gated by PCs or power-ups, gated by geographic layout, or just give player everything in a small sandbox and grow complexity of levels.
 - When possible though, try to teach players how to play through examples, visuals, etc.